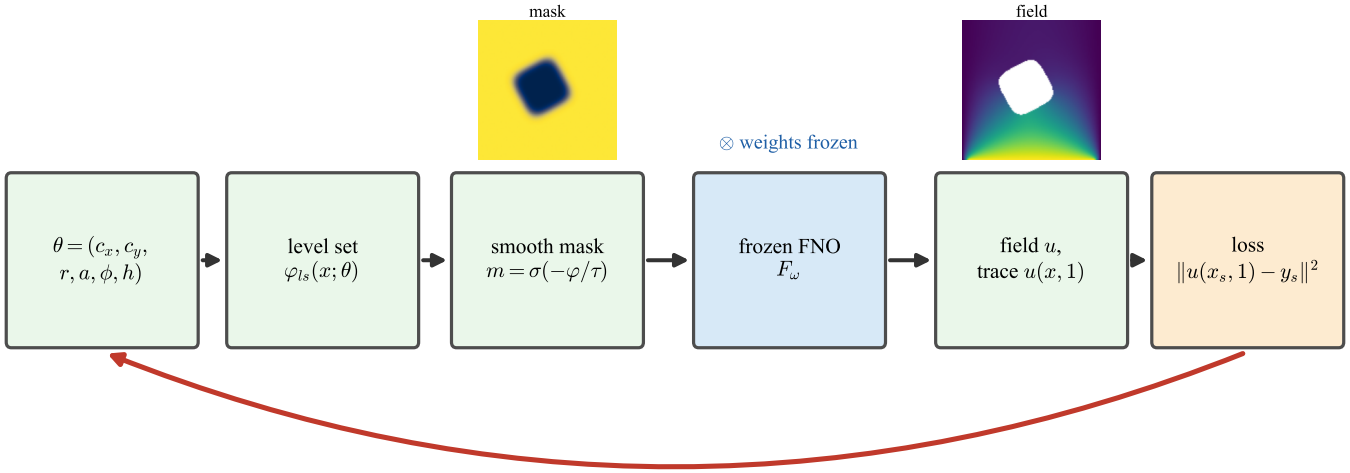

mask
field
⊗ weights frozen
$\theta = (c_x, c_y, r, a, \phi, h)$
level set $\varphi_{ls}(x;\theta)$
smooth mask $m = \sigma(-\varphi/\tau)$
frozen FNO F_ω
field u, trace $u(x,1)$
loss $\|u(x_s,1) - y_s\|^2$
autograd $\partial\mathcal{L}/\partial\theta$ (shape-derivative gradient, sourced on the $\mathcal{O}(\tau)$ boundary collar)